

Recitation 10

Q1)

Suppose the economy of a country is simplified to have only one good. The total annual production (nominal GDP) of this good in the current year is valued at \$2,000 billion. The average amount of money in circulation (the money supply) in the same year is \$500 billion.

1. Calculate the velocity of money in this economy.
2. Assume that next year the nominal GDP increases to \$2,200 billion, and the money supply increases to \$550 billion. What is the new velocity of money?
3. Explain briefly how the change in the velocity of money could impact inflation in the economy.

Q2)

1. Go to <https://fred.stlouisfed.org/> and search for the “effective federal funds rate” (EFFR). Alter the series so the graph displays the value of the EFFR from July 3, 2000 through April 3, 2023.
 - (a) How many recessions have there been over this time period? During those recessions, what happened to the EFFR? Provide an intuitive reason for why you are observing this trend/relationship.
 - (b) As of April 3, 2023, what is the value of the EFFR? _____ What was the value of the EFFR one year ago? _____ Recently, the Fed has been raising interest rates because it is concerned about inflation. Using the AD-AS model, describe the logic behind the Fed’s approach. Draw a graph to support your answer.

- (c) Suppose the Fed is successful in lowering the inflation rate. Explain why over time the inflation rate may fall even further. Show this change on the graph that you drew for part (b).
2. In the wake of the 1973 oil crisis and the 1979 energy crisis, stagflation (i.e., an increase in inflation coupled with a recession) began to afflict the U.S. economy. What kind of shock caused stagflation? Real or AD? Positive or negative? Show this impact on the graph below and identify the new inflation rate (11%) and real growth rate (-2%).

In response to the shocks described above, the Federal Reserve employed expansionary monetary policy throughout the 1970s to promote real economic growth. Although the Fed's policy promoted real economic growth, it made the inflation problem worse. Using the graph above, show how the Fed's policy increased inflation to 13% and increased the real growth rate to 6.6%.

3. In 1979, the U.S. inflation rate was over 11%. As you know from question #2, the increase in inflation was due to a combination of events, including the expansionary policy of the Federal Reserve.
- (a) Model the U.S. economy in the late 1970s by drawing an AD-LRAS-SRAS graph such that the actual inflation rate and expected inflation rate are both equal to 11.5% and real GDP growth is 0%.
- (b) In 1979, Paul Volcker was appointed chair of the Federal Reserve Board. He is widely credited with ending the high levels of inflation that the U.S. experienced during the 1970s and early 1980s by employing *contractionary monetary policy*. Given what you learned in class, what impact do you think Volcker's policy had on real economic growth in the U.S.? Show the impact that Volcker's policy had on real GDP growth in the graph that you drew above.
- (c) What impact do you think Volcker's monetary policy had on inflation expectations and actual inflation in the early 1980s? Use the graph that you drew in parts (a) and (b) to illustrate your answer.